

Foster
Summary of Product Characteristics

1. FOSTER 100/6 MCG/DOSE PRESSURISED INHALATION SOLUTIONNAME OF THE MEDICINAL PRODUCT

Foster 100/6 micrograms per actuation pressurised inhalation solution.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each metered dose (ex-valve) contains:

100 micrograms of beclometasone dipropionate and 6 micrograms of formoterol fumarate dihydrate. This is equivalent to a delivered dose (ex-actuator) of 84.6 micrograms of beclometasone dipropionate and 5.0 micrograms of formoterol fumarate dihydrate.

For a full list of excipients see section 6.1.

3. PHARMACEUTICAL FORM

Pressurised inhalation solution.

4. CLINICAL PARTICULARS

4.1 THERAPEUTIC INDICATIONS

Asthma

Foster is indicated in the regular treatment of asthma where use of a combination product (inhaled corticosteroid and long-acting beta₂-agonist) is appropriate:

- patients not adequately controlled with inhaled corticosteroids and 'as needed' inhaled rapid-acting beta₂-agonist or
- patients already adequately controlled on both inhaled corticosteroids and long-acting beta₂-agonists.

COPD

Symptomatic treatment of patients with severe COPD (FEV₁ < 50% predicted normal) and a history of repeated exacerbations, who have significant symptoms despite regular therapy with long-acting bronchodilators.

4.2 POSOLOGY AND METHOD OF ADMINISTRATION

FOSTER is for inhalation use.

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Posology

ASTHMA

FOSTER is not intended for the initial management of asthma. The dosage of the components of Foster is individual and should be adjusted to the severity of the disease. This should be considered not only when treatment with combination products is initiated but also when the dose is adjusted. If an individual patient should require a combination of doses other than those available in the combination inhaler, appropriate doses of beta₂-agonists and/or corticosteroids by individual inhalers should be prescribed.

Beclometasone dipropionate in Foster is characterised by an extrafine particle size distribution which results in a more potent effect than formulations of beclometasone dipropionate with a non-extrafine particle size distribution (100 micrograms of beclometasone dipropionate extrafine in Foster are equivalent to 250 micrograms of beclometasone dipropionate in a non-extrafine formulation). Therefore the total daily dose of beclometasone dipropionate administered in Foster should be lower than the total daily dose of beclometasone dipropionate administered in a non-extrafine beclometasone dipropionate formulation.

This should be taken into consideration when a patient is transferred from a beclometasone dipropionate non-extrafine formulation to Foster; the dose of beclometasone dipropionate should be lower and will need to be adjusted to the individual needs of the patients.

There are two treatment approaches:

A. Maintenance therapy: Foster is taken as regular maintenance treatment with a separate as needed rapid-acting bronchodilator.

B. Maintenance and reliever therapy: Foster is taken as regular maintenance treatment and as needed in response to asthma symptoms.

A. Maintenance therapy

Patients should be advised to have their separate rapid-acting bronchodilator available for rescue use at all times.

Dose recommendations for adults 18 years and above:

One or two inhalations twice daily.

The maximum daily dose is 4 inhalations.

B. Maintenance and reliever therapy

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Patients take their daily maintenance dose of Foster and in addition take Foster as needed in response to asthma symptoms. Patients should be advised to always have Foster available for rescue use.

Foster maintenance and reliever therapy should especially be considered for patients with:

- not fully controlled asthma and in need of reliever medication
- asthma exacerbations in the past requiring medical intervention

Close monitoring for dose-related adverse effects is needed in patients who frequently take high numbers of Foster as-needed inhalations.

Dose recommendations for adults 18 years and above:

The recommended maintenance dose is 1 inhalation twice daily (one inhalation in the morning and one inhalation in the evening).

Patients should take 1 additional inhalation as needed in response to symptoms. If symptoms persist after a few minutes, an additional inhalation should be taken.

The maximum daily dose is 8 inhalations.

Patients requiring frequent use of rescue inhalations daily should be strongly recommended to seek medical advice. Their asthma should be reassessed and their maintenance therapy should be reconsidered.

Dose recommendations for children and adolescents under 18 years:

The safety and efficacy of Foster in children and adolescents under 18 years of age have not been established yet. No data are available with Foster in children under 12 years of age. Only limited data are available in adolescents between 12 and 17 years of age. Therefore, Foster is not recommended for children and adolescents under 18 years until further data become available.

Patients should be regularly reassessed by a doctor, so that the dosage of Foster remains optimal and is only changed on medical advice. The dose should be titrated to the lowest dose at which effective control of symptoms is maintained. When control of symptoms is maintained with the lowest recommended dosage, then the next step could include a test of inhaled corticosteroid alone.

Patients should be advised to take Foster every day even when asymptomatic.

COPD

Dose recommendations for adults 18 years and above:

Two inhalations twice daily.

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Special patient groups:

There is no need to adjust the dose in elderly patients. There are no data available for use of Foster in patients with hepatic or renal impairment (see section 5.2).

Method of administration to ensure proper administration of the drug, the patient should be shown how to use the inhaler correctly by a physician or other health professional. Correct use of the pressurised metered dose inhaler is essential in order that treatment is successful. The patient should be advised to read the Patient Information Leaflet carefully and follow the instructions for use as given in the Leaflet.

Before using the inhaler for the first time or if the inhaler has not been used for 14 days or more, one actuation should be released into the air in order to ensure that the inhaler is working properly.

Whenever possible patients should stand or sit in an upright position when inhaling from their inhaler.

The steps below should be followed:

1. Remove the protective cap from the mouthpiece and check that the mouthpiece is clean and free from dust and dirt or any other foreign objects.
2. Breathe out as slowly and deeply as possible.
3. Hold the canister vertically with its body upwards and put the lips around the mouthpiece. Do not bite the mouthpiece.
4. At the same time, breathe in slowly and deeply through the mouth. After starting to breathe in press down on the top of the inhaler to release one puff.
5. Hold the breath for as long as possible and, finally, remove the inhaler from the mouth and breathe out slowly. Do not breathe out into the inhaler. Should a further puff be needed, keep the inhaler in a vertical position for about half a minute and repeat steps 2 to 5.

After use, close with protective cap.

IMPORTANT: do not perform steps 2 to 5 too quickly.

If mist appears following inhalation, either from the inhaler or from the sides of the mouth, the procedure should be repeated from step 2.

For patients with weak hands, it may be easier to hold the inhaler with both hands. Therefore, the index fingers should be placed on the top of the inhaler canister and both thumbs on the base of the inhaler.

Patients should rinse their mouth or gargle with water or brush the teeth after inhaling (see section 4.4).

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Cleaning

Patients should be advised to read the Patient Information Leaflet carefully for cleaning instructions. For the regular cleaning of the inhaler, patients should remove the cap from the mouthpiece and wipe the outside and inside of the mouthpiece with a dry cloth. They should not use water or other liquids to clean the mouthpiece.

Patients who find it difficult to synchronise aerosol actuation with inspiration of breath, may use the AeroChamber Plus® spacer device. They should be advised by their doctor, pharmacist or a nurse in the proper use and care of their inhaler and spacer and their technique checked to ensure optimum delivery of the inhaled drug to the lungs. This may be obtained by the patients using the AeroChamber Plus® by one continuous slow and deep breath through the spacer, without any delay between actuation and inhalation.

4.3 CONTRAINDICATIONS

Hypersensitivity to beclometasone dipropionate, formoterol fumarate dihydrate or any of the excipients listed in section 6.1.

4.4 SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Pneumonia in Patient with COPD

An increase in the incidence of pneumonia, including pneumonia requiring hospitalization, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies.

There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products.

Physicians should remain vigilant for the possible development of pneumonia in patient with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations.

Risk factors for pneumonia in patients with COPD include current smoking status, older age, low body mass index (BMI) and severe COPD.

Foster should be used with caution (which may include monitoring) in patients with cardiac arrhythmias, especially third degree atrioventricular block and tachyarrhythmias (accelerated and/or irregular heart beat), idiopathic subvalvular aortic stenosis, hypertrophic obstructive cardiomyopathy, severe heart disease, particularly acute myocardial infarction, ischaemic heart disease, congestive

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heart failure, occlusive vascular diseases, particularly arteriosclerosis, arterial hypertension and aneurysm.

Caution should also be observed when treating patients with known or suspected prolongation of the QTc interval, either congenital or drug induced (QTc > 0.44 seconds). Formoterol itself may induce prolongation of the QTc interval.

Caution is also required when Foster is used by patients with thyrotoxicosis, diabetes mellitus, phaeochromocytoma and untreated hypokalaemia.

Potentially serious hypokalaemia may result from beta₂-agonist therapy. Particular caution is advised in severe asthma as this effect may be potentiated by hypoxia. Hypokalaemia may also be potentiated by concomitant treatment with other drugs which can induce hypokalaemia, such as xanthine derivatives, steroids and diuretics (see Section 4.5). Caution is also recommended in unstable asthma when a number of “rescue” bronchodilators may be used. It is recommended that serum potassium levels are monitored in such situations.

The inhalation of formoterol may cause a rise in blood glucose levels. Therefore, blood glucose should be closely monitored in patients with diabetes.

If anaesthesia with halogenated anaesthetics is planned, it should be ensured that Foster is not administered for at least 12 hours before the start of anaesthesia as there is a risk of cardiac arrhythmias.

As with all inhaled medication containing corticosteroids, Foster should be administered with caution in patients with active or quiescent pulmonary tuberculosis, fungal and viral infections in the airways.

It is recommended that treatment with Foster should not be stopped abruptly.

If patients find the treatment ineffective medical attention must be sought. Increasing use of rescue bronchodilators indicates a worsening of the underlying condition and warrants a reassessment of the asthma therapy. Sudden and progressive deterioration in control of asthma or COPD is potentially life-threatening and the patient should undergo urgent medical assessment. Consideration should be given to the need for increased treatment with corticosteroids, either inhaled or oral therapy, or antibiotic treatment if an infection is suspected.

Patients should not be initiated on Foster during an exacerbation, or if they have significantly worsening or acutely deteriorating asthma. Serious asthma-related adverse events and exacerbations may occur during treatment with Foster. Patients should be asked to continue treatment but to seek medical advice if asthma symptoms remain uncontrolled or worsen after initiation on Foster.

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As with other inhalation therapy paradoxical bronchospasm may occur with an immediate increase in wheezing and shortness of breath after dosing. This should be treated immediately with a fast-acting inhaled bronchodilator. Foster should be discontinued immediately, the patient assessed, and alternative therapy instituted if necessary.

Foster should not be used as the first treatment for asthma.

For treatment of acute asthma attacks patients should be advised to have their rapid-acting bronchodilator available at all times, either Foster (for patients using Foster as maintenance and reliever therapy) or a separate rapid-acting bronchodilator (for patients using Foster as maintenance therapy only).

Patients should be reminded to take Foster daily as prescribed even when asymptomatic. The reliever inhalations of Foster should be taken in response to asthma symptoms but are not intended for regular prophylactic use, e.g. before exercise. For such use, a separate rapid-acting bronchodilator should be considered.

Once asthma symptoms are controlled, consideration may be given to gradually reducing the dose of Foster. Regular review of patients as treatment is stepped down is important. The lowest effective dose of Foster should be used (see section 4.2).

Systemic effects may occur with any inhaled corticosteroid, particularly at high doses prescribed for long periods. These effects are much less likely to occur with inhaled than with oral corticosteroids. Possible systemic effects include: Cushing's syndrome, Cushingoid features, adrenal suppression, decrease in bone mineral density, growth retardation in children and adolescents, cataract and glaucoma and more rarely, a range of psychological or behavioural effects including psychomotor hyperactivity, sleep disorders, anxiety, depression or aggression (particularly in children).

Therefore, it is important that the patient is reviewed regularly, and the dose of inhaled corticosteroid is reduced to the lowest dose at which effective control of asthma is maintained.

Single dose pharmacokinetic data (see section 5.2) have demonstrated that the use of Foster with Aerochamber Plus® spacer device in comparison to the use of standard actuator, does not increase the total systemic exposure to formoterol and reduces the systemic exposure to beclometasone-17-monopropionate, while there is an increase for unchanged beclometasone dipropionate that reaches systemic circulation from the lung; however, since the total systemic exposure to beclometasone dipropionate plus its active metabolite does not change, there is no increased risk of systemic effects when using Foster with the named spacer device.

Prolonged treatment of patients with high doses of inhaled corticosteroids may result in adrenal suppression and acute adrenal crisis. Children aged less than 16 years taking/inhaling higher than recommended doses of beclometasone dipropionate may be at particular risk. Situations which could potentially trigger acute adrenal crisis, include trauma, surgery, infection or any rapid reduction in

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dosage. Presenting symptoms are typically vague and may include anorexia, abdominal pain, weight loss, tiredness, headache, nausea, vomiting, hypotension, decreased level of consciousness, hypoglycaemia, and seizures. Additional systemic corticosteroid cover should be considered during periods of stress or elective surgery.

Care should be taken when transferring patients to Foster therapy, particularly if there is any reason to suppose that adrenal function is impaired from previous systemic steroid therapy.

Patients transferring from oral to inhaled corticosteroids may remain at risk of impaired adrenal reserve for a considerable time. Patients who have required high dose emergency corticosteroid therapy in the past or have received prolonged treatment with or high doses of inhaled corticosteroids may also be at risk. This possibility of residual impairment should always be borne in mind in emergency and elective situations likely to produce stress, and appropriate corticosteroid treatment must be considered. The extent of the adrenal impairment may require specialist advice before elective procedures.

Patients should be advised that Foster contains a small amount of ethanol (approximately 7 mg per actuation); however, at normal doses the amount of ethanol is negligible and does not pose a risk to patients.

Patients should be advised to rinse the mouth or gargle with water or brush the teeth after inhaling the prescribed dose to minimise the risk of oropharyngeal candida infection.

4.5 INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Pharmacokinetic interactions

Beclometasone dipropionate undergoes a very rapid metabolism via esterase enzymes without involvement of cytochrome P450 system.

Beclomethasone is less dependent on CYP3A metabolism than some other corticosteroids, and in general interaction are unlikely; however the possibility of systemic effects with concomitant use of strong CYP3A inhibitors (e.g. Cobicistat) cannot be excluded, and therefore caution and appropriate monitoring is advised with the use of such agents.

Pharmacodynamic interactions

Beta-blockers (including eye drops) should be avoided in asthmatic patients. If beta-blockers are administered for compelling reasons, the effect of formoterol will be reduced or abolished. On the other hand, concomitant use of other beta-adrenergic drugs can have potentially additive effects, therefore caution is required when theophylline or other beta-adrenergic drugs are prescribed concomitantly with formoterol.

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Concomitant treatment with quinidine, disopyramide, procainamide, phenothiazines, antihistamines, monoamine oxidase inhibitors and tricyclic antidepressants can prolong the QTc-interval and increase the risk of ventricular arrhythmias.

In addition L-dopa, L-thyroxine, oxytocin and alcohol can impair cardiac tolerance towards beta₂-sympathomimetics.

Concomitant treatment with monoamine oxidase inhibitors including agents with similar properties such as furazolidone and procarbazine may precipitate hypertensive reactions. There is an elevated risk of arrhythmias in patients receiving concomitant anaesthesia with halogenated hydrocarbons.

Concomitant treatment with xanthine derivatives, steroids, or diuretics may potentiate a possible hypokalaemic effect of beta₂-agonists (see section 4.4.). Hypokalaemia may increase the disposition towards arrhythmias in patients who are treated with digitalis glycosides.

Foster contains a small amount of ethanol. There is a theoretical potential for interaction in particularly sensitive patients taking disulfiram or metronidazole

4.6 FERTILITY, PREGNANCY AND LACTATION

There is no experience with or evidence of safety of propellant HFA-134a in human pregnancy or lactation. However, studies of the effect of HFA-134a on reproductive function and embryofetal development in animals have revealed no clinically relevant adverse effects.

Pregnancy

There are no relevant clinical data on the use of Foster in pregnant women. Animal studies using beclometasone dipropionate and formoterol combination showed evidence of toxicity to reproduction after high systemic exposure (see 5.3 Preclinical safety data). Because of the tocolytic actions of beta₂-sympathomimetic agents particular care should be exercised in the run up to delivery. Formoterol should not be recommended for use during pregnancy and particularly at the end of pregnancy or during labour unless there is no other (safer) established alternative.

Foster should only be used during pregnancy if the expected benefits outweigh the potential risks.

Lactation

There are no relevant clinical data on the use of Foster in lactation in humans.

Although no data from animal experiments are available, it is reasonable to assume that beclometasone dipropionate is secreted in milk, like other corticosteroids.

While it is not known whether formoterol passes into human breast milk, it has been detected in the milk of lactating animals.

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Administration of Foster to women who are breast-feeding should only be considered if the expected benefits outweigh the potential risks.

4.7 EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Foster is unlikely to have any effect on the ability to drive and operate machinery.

4.8 UNDESIRABLE EFFECTS

As Foster contains beclometasone dipropionate and formoterol fumarate dihydrate, the type and severity of adverse reactions associated with each of the compounds may be expected. There is no incidence of additional adverse events following concurrent administration of the two compounds.

Undesirable effects which have been associated with beclometasone dipropionate and formoterol administered as a fixed combination (Foster) and as single agents are given below, listed by system organ class.

Frequencies are defined as:

very common ($\geq 1/10$),

common ($\geq 1/100$ and $< 1/10$),

uncommon ($\geq 1/1,000$ and $< 1/100$), rare ($\geq 1/10,000$ and $< 1/1,000$) and very rare ($\leq 1/10,000$).

Common and uncommon ADRs were derived clinical trials in asthmatic and COPD patients.

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System Organ Class	Adverse Reaction	Frequency
Infections and Infestations	Pharyngitis, oral candidiasis, Pneumonia (in COPD patients)	Common
	Influenza, oral fungal infection, pharyngeal and oesophageal candidiasis, vaginal candidiasis, gastroenteritis, sinusitis, rhinitis, pneumonia*	Uncommon
Blood and the lymphatic system disorders	Granulocytopenia	Uncommon
	Thrombocytopenia	Very rare
Immune system disorders	Dermatitis allergic	Uncommon
	Hypersensitivity reactions, including erythema, lips, face, eye and pharyngeal oedema	Very rare
Endocrine disorders	Adrenal suppression	Very rare
Metabolism and nutrition disorders	Hypokalaemia, hyperglycaemia	Uncommon
Psychiatric disorders	Restlessness	Uncommon
	Psychomotor hyperactivity, sleep disorders, anxiety, depression, aggression, behavioural changes (predominantly in children)	Unknown
Nervous system disorders	Headache	Common
	Tremor, dizziness,	Uncommon
Eye disorders	Glaucoma, cataract	Very rare
Ear and labyrinth disorders	Otosalpingitis	Uncommon
Cardiac disorders	Palpitations, electrocardiogram QT corrected interval prolonged, electrocardiogram change, tachycardia, tachyarrhythmia, atrial fibrillation*	Uncommon
	Ventricular extrasystoles, angina pectoris	Rare

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Vascular disorders	Hyperaemia, flushing	Uncommon
Respiratory, thoracic and mediastinal disorders	Dysphonia	Common
	Cough, productive cough, throat irritation, asthmatic crisis	Uncommon
	Bronchospasm paradoxical	Rare
	Dyspnoea, exacerbation of asthma	Very rare
Gastrointestinal disorders	Diarrhoea, dry mouth, dyspepsia, dysphagia, burning sensation of the lips, nausea, dysgeusia	Uncommon
Skin and subcutaneous tissue disorders	Pruritus, rash, hyperhidrosis, urticaria,	Uncommon
	angioneurotic oedema	Rare
Musculoskeletal, connective tissue and bone disorders	Muscle spasms, myalgia	Uncommon
	Growth retardation in children and adolescents	Very rare
Renal and urinary disorders	Nephritis	Rare
General disorders and administration site conditions	Oedema peripheral	Very rare
Investigations	C-reactive protein increased, platelet count increased, free fatty acids increased, blood insulin increased, blood ketone body increased, blood cortisol decrease*	Uncommon
	Blood pressure increased, blood pressure decreased	Rare
	Bone density decreased	Very rare

* One related non serious case of pneumonia was reported by one patient treated with FOSTER in a pivotal clinical trial in COPD patients. Other adverse reactions observed with FOSTER in COPD clinical trials were: reduction of blood cortisol and atrial fibrillation.

As with other inhalation therapy, paradoxical bronchospasm may occur (see 4.4 'Special Warnings and Precautions for Use').

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Among the observed adverse reactions those typically associated with formoterol are:

hypokalaemia, headache, tremor, palpitations, cough, muscle spasms and prolongation of QTc interval.

Adverse reactions typically associated with the administration of beclometasone dipropionate are: oral fungal infections, oral candidiasis, dysphonia, throat irritation.

Dysphonia and candidiasis may be relieved by gargling or rinsing the mouth with water or brushing the teeth after using the product. Symptomatic candidiasis can be treated with topical anti-fungal therapy whilst continuing the treatment with Foster.

Systemic effects of inhaled corticosteroids (e.g. beclometasone dipropionate) may occur particularly when administered at high doses prescribed for prolonged periods, these may include adrenal suppression, decrease in bone mineral density, growth retardation in children and adolescents, cataract and glaucoma (see also 4.4).

Hypersensitivity reactions including rash, urticaria pruritus, erythema and oedema of the eyes, face, lips and throat may also occur.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 OVERDOSE

Inhaled doses of Foster up to twelve cumulative actuations (total beclometasone dipropionate 1200 micrograms, Formoterol 72 micrograms) have been studied in asthmatic patients. The cumulative treatments did not cause abnormal effect on vital signs and neither serious nor severe adverse events were observed.

Excessive doses of formoterol may lead to effects that are typical of beta₂-adrenergic agonists: nausea, vomiting, headache, tremor, somnolence, palpitations, tachycardia, ventricular arrhythmias, prolongation of QTc interval, metabolic acidosis, hypokalaemia, hyperglycaemia.

In case of overdose of formoterol, supportive and symptomatic treatment is indicated. Serious cases should be hospitalised. Use of cardioselective beta-adrenergic blockers may be considered, but only subject to extreme caution since the use of beta-adrenergic blocker medication may provoke bronchospasm. Serum potassium should be monitored.

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Acute inhalation of beclometasone dipropionate doses in excess of those recommended may lead to temporary suppression of adrenal function. This does not need emergency action as adrenal function recovers in a few days, as verified by plasma cortisol measurements. In these patients treatment should be continued at a dose sufficient to control asthma.

Chronic overdose of inhaled beclometasone dipropionate: risk of adrenal suppression (see section 4.4.). Monitoring of adrenal reserve may be necessary. Treatment should be continued at a dose sufficient to control asthma.

5 PHARMACOLOGICAL PROPERTIES

5.1 PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Adrenergics and other drugs for obstructive airway diseases.

ATC-code: R03 AK08.

Mechanisms of action and pharmacodynamic effects

Foster contains beclometasone dipropionate and formoterol. These two actives have different modes of action. In common with other inhaled corticosteroids and beta₂-agonist combinations, additive effects are seen in respect of reduction in asthma exacerbations.

Beclometasone dipropionate

Beclometasone dipropionate given by inhalation at recommended doses has a glucocorticoid antiinflammatory action within the lungs, resulting in reduced symptoms and exacerbations of asthma with less adverse effects than when corticosteroids are administered systemically.

Formoterol

Formoterol is a selective beta₂-adrenergic agonist that produces relaxation of bronchial smooth muscle in patients with reversible airways obstruction. The bronchodilating effect sets in rapidly, within 1-3 minutes after inhalation, and has a duration of 12 hours after a single dose.

ASTHMA

Clinical efficacy for Foster maintenance therapy

In clinical trials in adults, the addition of formoterol to beclometasone dipropionate improved asthma symptoms and lung function and reduced exacerbations.

In a 24-week study the effect on lung function of Foster was at least equal to that of the free combination of beclometasone dipropionate and formoterol, and exceeded that of beclometasone dipropionate alone.

Clinical efficacy for Foster maintenance and reliever therapy

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In a 48-week parallel group study involving 1701 asthma patients, the efficacy of Foster administered as maintenance (1 inhalation BID) and reliever therapy (up to a total of 8 puffs per day) was compared to Foster administered as maintenance therapy (1 inhalation BID) plus as needed salbutamol, in adult patients with un-controlled moderate to severe asthma. The results demonstrated that Foster used as maintenance and reliever therapy significantly prolonged the time to first severe exacerbation (*) when compared with Foster used as maintenance plus as needed salbutamol ($p < 0.001$ for both ITT and PP population). The rate of severe asthma exacerbations per patients/year, was significantly reduced in the maintenance and reliever therapy group compared to salbutamol group: 0.1476 vs 0.2239 respectively (statistically significant reduction: $p < 0.001$). Patients in the Foster maintenance and reliever group achieved a clinically meaningful improvement in asthma control. The mean number of inhalations/day of reliever medication and the proportion of patients using reliever medication decreased similarly in both groups.

Note*: severe exacerbations were defined as deterioration in asthma resulting in hospitalisation or emergency room treatment, or resulting in the need for systemic steroids for more than 3 days

In another clinical study, a single dose of **FOSTER 100/6 mcg** provided a quick bronchodilation effect and a rapid relief from dyspnea symptoms similar to that of salbutamol 200 mcg/dose in asthmatic patients when metacholine challenge is used to induce bronchoconstriction.

COPD

In two 48-weeks studies, the effects on lung function and the rate of exacerbation (defined as courses of oral steroids and/or course of antibiotics and/or hospitalisations) in patients with severe COPD (30% $< FEV_1 < 50\%$) was evaluated.

One pivotal trial showed a significant improvement in lung function (primary endpoint change in pre-dose FEV_1) compared to formoterol after 12 weeks of treatment (adjusted mean difference between Foster and formoterol: 69 ml) as well as at each clinic visit during the whole treatment period (48 weeks). The study demonstrated that the mean number of exacerbations per patient/year (exacerbation rate, co-primary endpoint) was statistically significantly reduced with Foster as compared with formoterol treatment (adjusted mean rate 0.80 compared with 1.12 in the formoterol group, adjusted ratio 0.72, $p < 0.001$) over 48 weeks treatment period in a total of 1199 patients with severe COPD. In addition, Foster statistically significantly prolonged the time to first exacerbation compared to formoterol. The superiority of Foster versus formoterol was also confirmed in terms of exacerbation rate in subgroups of patients taking (around 50% in each treatment arm) or not Tiotropium Bromide as concomitant medication.

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The other pivotal study, which was a three arm, randomised, parallel group study in 718 patients, confirmed the superiority of Foster versus formoterol treatment in terms of change in pre-dose FEV₁ at the end of treatment (48 weeks) and demonstrated the non-inferiority of Foster compared to budesonide/formoterol fixed dose combination on the same parameter.

5.2 PHARMACOKINETIC PROPERTIES

The systemic exposure to the active substances beclometasone dipropionate and formoterol in the fixed combination Foster have been compared to the single components.

In a pharmacokinetic study conducted in healthy subjects treated with a single dose of Foster fixed combination (4 puffs of 100/6 micrograms) or a single dose of beclometasone dipropionate CFC (4 puffs of 250 micrograms) and Formoterol HFA (4 puffs of 6 micrograms), the AUC of beclometasone dipropionate main active metabolite (beclometasone-17-monopropionate) and its maximal plasma concentration were, respectively, 35% and 19% lower with the fixed combination than with non-extrafine beclometasone dipropionate CFC formulation, in contrast, the rate of absorption was more rapid (0.5 vs 2h) with the fixed combination compared to non-extrafine beclometasone dipropionate CFC formulation alone.

For formoterol, maximal plasma concentration was similar after administration of the fixed or the extemporaneous combination and the systemic exposure was slightly higher after administration of Foster than with the extemporaneous combination.

There was no evidence of pharmacokinetic or pharmacodynamic (systemic) interactions between beclometasone dipropionate and formoterol.

The use of Aerochamber Plus® spacer increased the lung delivery of beclometasone dipropionate active metabolite beclometasone 17-monopropionate and formoterol of 41% and 45% respectively, in comparison to the use of standard actuator in a study conducted in healthy volunteers. The total systemic exposure was unchanged for formoterol, reduced by 10% for beclometasone 17-monopropionate and increased for unchanged beclometasone dipropionate.

A lung deposition study conducted in stable COPD patients, healthy volunteers and asthmatic patients, demonstrated that on average 33% of the nominal dose is deposited into the lung of COPD patients compared to 34% in healthy subjects and 31% in asthmatic patients. Beclometasone 17-monopropionate and formoterol plasma exposures were comparable across the three groups during the 24 hours following the inhalation. The total exposure of beclometasone dipropionate was higher in COPD patients compared to the exposure in asthmatic patients and healthy volunteers.

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Beclometasone dipropionate

Beclometasone dipropionate is a pro-drug with weak glucocorticoid receptor binding affinity that is hydrolysed via esterase enzymes to an active metabolite beclometasone-17-monopropionate which has a more potent topical anti-inflammatory activity compared with the pro-drug beclometasone dipropionate.

Absorption, distribution and biotransformation

Inhaled beclometasone dipropionate is rapidly absorbed through the lungs; prior to absorption there is extensive conversion to its active metabolite beclometasone-17-monopropionate via esterase enzymes that are found in most tissues. The systemic availability of the active metabolite arises from lung (36 %) and from gastrointestinal absorption of the swallowed dose. The bioavailability of swallowed beclometasone dipropionate is negligible however, pre-systemic conversion to beclometasone-17-monopropionate results in 41% of the dose being absorbed as the active metabolite.

There is an approximately linear increase in systemic exposure with increasing inhaled dose.

The absolute bioavailability following inhalation is approximately 2% and 62% of the nominal dose for unchanged beclometasone dipropionate and beclometasone-17-monopropionate respectively.

Following intravenous dosing, the disposition of beclometasone dipropionate and its active metabolite are characterised by high plasma clearance (150 and 120L/h respectively), with a small volume of distribution at steady state for beclometasone dipropionate (20L) and larger tissue distribution for its active metabolite (424L).

Plasma protein binding is moderately high.

Elimination

Faecal excretion is the major route of beclometasone dipropionate elimination mainly as polar metabolites. The renal excretion of beclometasone dipropionate and its metabolites is negligible. The terminal elimination half-lives are 0.5 h and 2.7 h for beclometasone dipropionate and beclometasone-17-monopropionate respectively.

Special populations

The pharmacokinetics of beclometasone dipropionate in patients with renal or hepatic impairment has not been studied; however, as beclometasone dipropionate undergoes a very rapid metabolism via

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esterase enzymes present in intestinal fluid, serum, lungs and liver, to originate the more polar products beclometasone-21-monopropionate, beclometasone-17-monopropionate and beclometasone, hepatic impairment is not expected to modify the pharmacokinetics and safety profile of beclometasone dipropionate.

As beclometasone dipropionate or its metabolites were not traced in the urine, an increase in systemic exposure is not envisaged in patients with renal impairment.

Formoterol

Absorption and distribution

Following inhalation, formoterol is absorbed both from the lung and from the gastrointestinal tract. The fraction of an inhaled dose that is swallowed after administration with a metered dose inhaler (MDI) may range between 60% and 90%. At least 65% of the fraction that is swallowed is absorbed from the gastrointestinal tract. Peak plasma concentrations of unchanged drug occur within 0.5 to 1 hours after oral administration. Plasma protein binding of formoterol is 61-64% with 34% bound to albumin. There was no saturation of binding in the concentration range attained with therapeutic doses. The elimination half-life determined after oral administration is 2-3 hours. Absorption of formoterol is linear following inhalation of 12 to 96 µg of formoterol fumarate.

Biotransformation

Formoterol is widely metabolised and the prominent pathway involves direct conjugation at the phenolic hydroxyl group. Glucuronide acid conjugate is inactive. The second major pathway involves O-demethylation followed by conjugation at the phenolic 2'-hydroxyl group. Cytochrome P450 isoenzymes CYP2D6, CYP2C19 and CYP2C9 are involved in the O-demethylation of formoterol. Liver appears to be the primary site of metabolism. Formoterol does not inhibit CYP450 enzymes at therapeutically relevant concentrations.

Elimination

The cumulative urinary excretion of formoterol after single inhalation from a dry powder inhaler increased linearly in the 12 – 96 µg dose range. On average, 8% and 25% of the dose was excreted as unchanged and total formoterol, respectively. Based on plasma concentrations measured following inhalation of a single 120 µg dose by 12 healthy subjects, the mean terminal elimination half-life was determined to be 10 hours. The (R,R)- and (S,S)-enantiomers represented about 40% and 60% of unchanged drug excreted in the urine, respectively. The relative proportion of the two enantiomers remained constant over the dose range studied and there was no evidence of relative accumulation of one enantiomer over the other after repeated dosing.

After oral administration (40 to 80 µg), 6% to 10% of the dose was recovered in urine as unchanged drug in healthy subjects; up to 8% of the dose was recovered as the glucuronide.

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A total 67% of an oral dose of formoterol is excreted in urine (mainly as metabolites) and the remainder in the faeces. The renal clearance of formoterol is 150 ml/min.

Special populations

Hepatic/Renal impairment: the pharmacokinetics of formoterol has not been studied in patients with hepatic or renal impairment. However, as formoterol is primarily eliminated via hepatic metabolism, an increased exposure can be expected in patients with severe liver cirrhosis.

5.3 PRECLINICAL SAFETY DATA

The toxicity observed in animal studies with beclometasone dipropionate and formoterol, given in combination or separately, consisted mainly of effects associated with exaggerated pharmacological activity. They are related to the immuno-suppressive activity of beclometasone dipropionate and to the known cardiovascular effects of formoterol evident mainly in dogs. Neither increase in toxicity nor occurrence of unexpected findings were observed upon administration of the combination.

Reproduction studies in rats showed dose-dependent effects. The combination was associated with reduced female fertility and embryofetal toxicity. High doses of corticosteroids to pregnant animals are known to cause abnormalities of fetal development including cleft palate and intra-uterine growth retardation, and it is likely that the effects seen with the beclometasone dipropionate /formoterol combination were due to beclometasone dipropionate. These effects were noted only with high systemic exposure to the active metabolite beclometasone-17-monopropionate (200 fold the expected plasma levels in patients). Additionally, increased duration of gestation and parturition, an effect attributable to the known tocolytic effects of beta₂-sympathomimetics, was seen in animal studies. These effects were noted when maternal plasma formoterol levels were below the levels expected in patients treated with Foster.

Genotoxicity studies performed with a beclometasone dipropionate/formoterol combination do not indicate mutagenic potential. No carcinogenicity studies have been performed with the proposed combination. However animal data reported for the individual constituents do not suggest any potential risk of carcinogenicity in man.

Pre-clinical data on the CFC-free propellant HFA-134a reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential and toxicity to reproduction.

6 PHARMACEUTICAL PARTICULARS

6.1 LIST OF EXCIPIENTS

Norflurane (HFA-134a), Ethanol anhydrous, Hydrochloric acid

6.2 INCOMPATIBILITIES

Not applicable.

6.3 SHELF LIFE

17 months.

6.4 SPECIAL PRECAUTIONS FOR STORAGE

Prior to dispensing to the patient:

Store in a refrigerator (2-8°C) (for a maximum of 15 months).

After dispensing:

Do not store above 30°C (for a maximum of 2 months).

The canister contains a pressurised liquid. Do not expose to temperatures higher than 50°C. Do not pierce the canister.

6.5 NATURE AND CONTENTS OF CONTAINER

The inhalation solution is contained in a pressurised aluminium container sealed with a metering valve and fitted into a polypropylene plastic actuator which incorporates a mouthpiece and is provided with a plastic protective cap.

Each pack contains:

1 pressurised container which provides 120 actuations

6.6 SPECIAL PRECAUTIONS FOR DISPOSAL AND OTHER HANDLING

For pharmacies:

Enter the date of dispensing to the patient on the pack.

Ensure that there is a period of at least 2 months between the date of dispensing and the expiry date printed on the pack.

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7 PRODUCT REGISTRATION HOLDER

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Selangor

8 DATE OF REVISION OF THE TEXT: February 2024